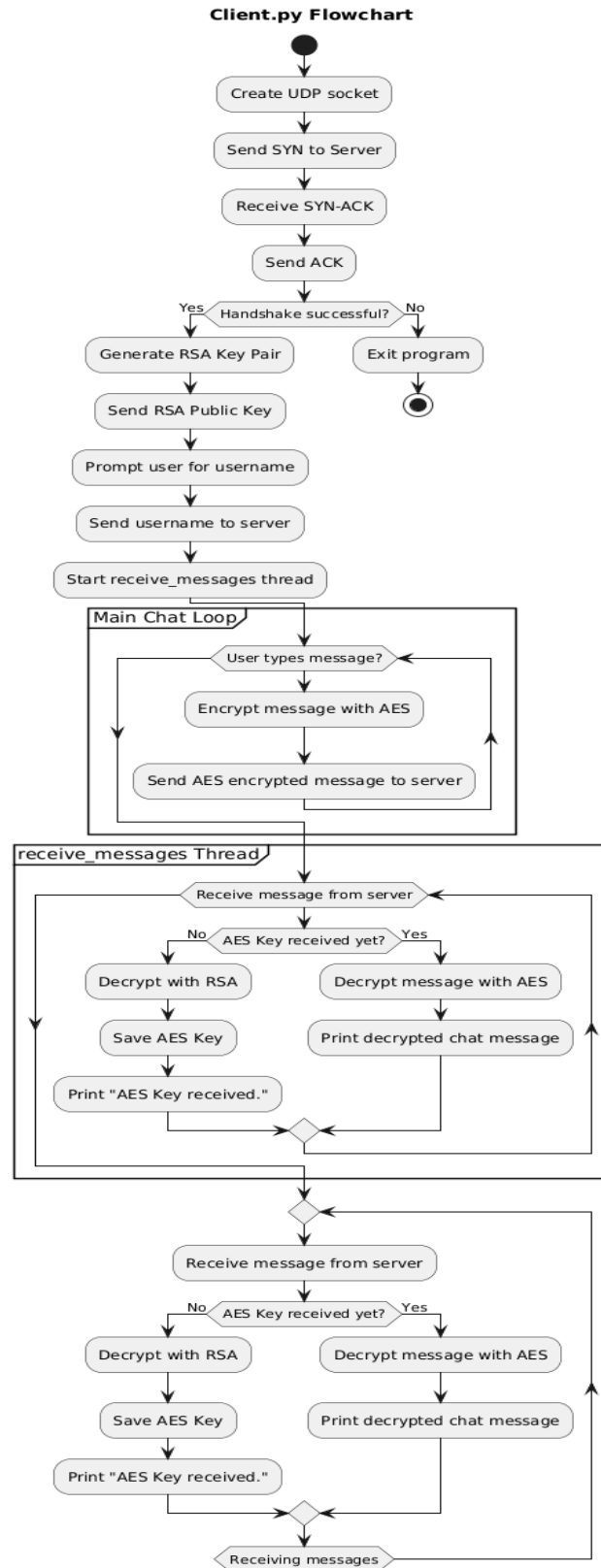
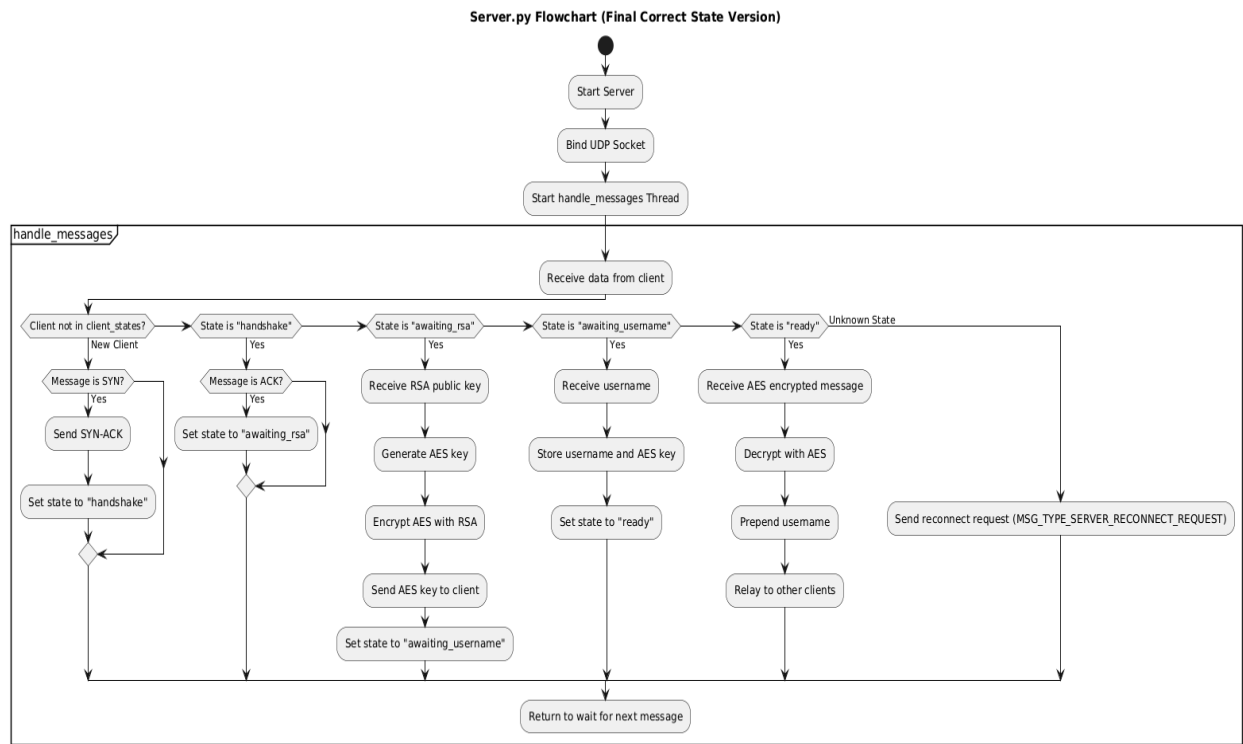




Client.py flowchart

Server.py flowchart



Message Handling Section:

Client to Server

Client to Server (Incoming Message)				
Size	Field	Description	Default / Initial Value	
1 byte	Type	Message type (SYN, ACK, RSA Key Exchange, Username, AES Encrypted Chat)	0 (SYN) for first connection	
32 bytes	Username	Username of the sender (padded to 32 bytes if necessary)	"" (empty) until username sent	
Variable	Payload	Plaintext (RSA key or Username) or AES Encrypted Chat Message	"" (empty string)	

Server to Client

Server to Client				
Size	Field	Description	Default / Initial Value	
1 byte	Type	Message type (SYN-ACK, ACK, RSA Key Exchange Response, Username Confirm, AES Encrypted Chat)	1 (SYN-ACK) for first response	
32 bytes	Username	Sender username (padded to 32 bytes)	"" (empty) until username known	
Variable	Payload	Plaintext (AES key during key exchange) or AES Encrypted Chat Message	"" (empty string)	

Reconnect Request:

Size	Field	Description	Default / Initial Value
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1 byte	Type	(Server Special) Reconnect Request	~"SERVER_RECONNECT_REQUEST"~

Sequence Diagram:

